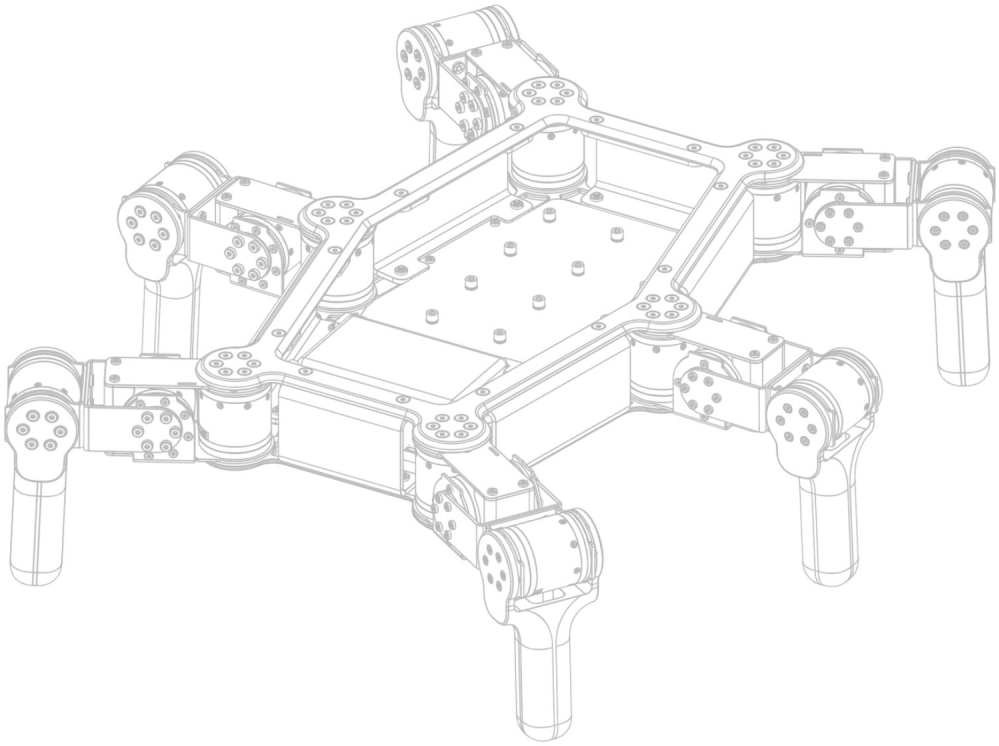
Cornell Physical Intelligence

Official Sponsorship Packet
2026–2027



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About Us

Cornell Physical Intelligence (CUPI) is a student robotics organization at Cornell University, founded in fall 2025. We design and build robots that can perceive their surroundings, make decisions, and act in the physical world.

Much of the recent progress in AI has happened in software. Our focus is on bringing that progress onto real hardware, which means treating mechanical design, electronics, and learning-based control as parts of one problem. Our projects cover manipulation, perception, and navigation on both ground and aerial robots, and each one is developed and tested on physical systems rather than in simulation alone.

Our members are curious, take initiative, and feel comfortable working on problems without an answer.



CUPI has 30 members across four subteams: mechanical, electrical, software, and business and marketing. Together, they take each project through design, fabrication, and testing, from custom circuit boards and machined parts to the software that runs on board.

Our main focus this year is Hexapod MKII, a six-legged metal walking robot with custom electronics and a walking policy trained in simulation. We are also training manipulation policies on a robotic arm, and this fall we will co-host the Physical Hackathon with the Cornell Entrepreneurship Club.

We keep a public wiki for all of our work. It records design reviews, test results, and code reviews, so anyone on or outside the team can follow how decisions were made.
It is available at: wiki.cornellphysicalintelligence.com.

What We Build



Hexapod MKII

Our flagship build and the continuation of last semester's walker. Hexapod MKII is a six-legged robot with 18 quasi-direct-drive joints, weighing 5–10 kg, with an all-metal structure and custom PCBs throughout. Its gait is trained with reinforcement learning (PPO with motion priors) in Isaac Lab on a DGX Spark, then runs onboard on a Jetson Orin, with a Livox Mid-360 lidar for perception. Built in collaboration with CU GeoData.



Robotic Arm

The robot arm project allows our team to explore different physical AI models while having a consistent and well documented hardware platform to work on. Our software subteam uses it experimentally to test and tweak vision robotics models, and explore what is possible with physical AI.



Autonomous Drone Racing

We compete in Anduril's AI Grand Prix, run with the Drone Champions League. Our team passed Virtual Qualifier 1 with a fully deterministic policy, clearing every gate with no learned network. Virtual Qualifier 2 removes pose and gate telemetry, so our policy now guides on bearings alone, from a single monocular camera and an IMU, using optical looming to judge distance to each gate. Both approaches are written up below.

TECHNICAL REPORTS



Passing Every Gate Without Learning

cornellphysicalintelligence.com/work/vq1-deterministic-policy



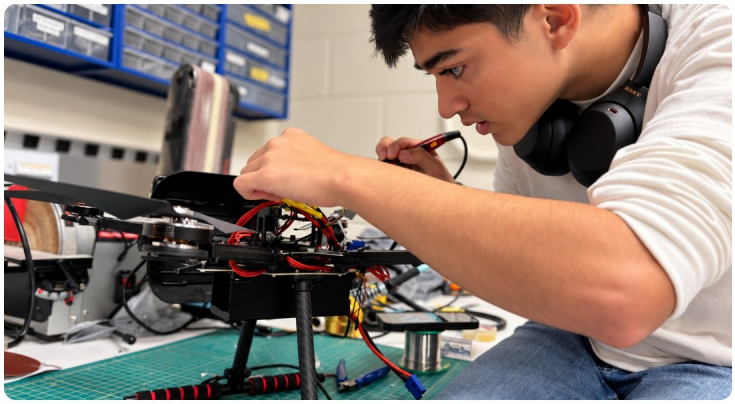
Racing Without a Map

cornellphysicalintelligence.com/work/racing-without-a-map

Our Subteams

Mechanical designs the structure and body of each robot; we create the chassis of our club's intelligent systems. Students mostly work with 3D printed plastics and metals, milled metals, and plasma cut parts.





Electrical handles power distribution and coordinates the flow of information between each of the robot's components, including sensors, servos, batteries, and onboard computers. Students design custom printed circuit boards, implement digital logic on FPGAs, develop embedded software for onboard computing systems, and test power distribution systems.

Software builds the stack that lets the robots we build interpret their environments and act inside them. Students work on computer vision, navigation, and control systems, building in simulation and testing on physical robots. They also build pipelines that use recorded sensor data to train vision models, tune controllers, and trace failures.

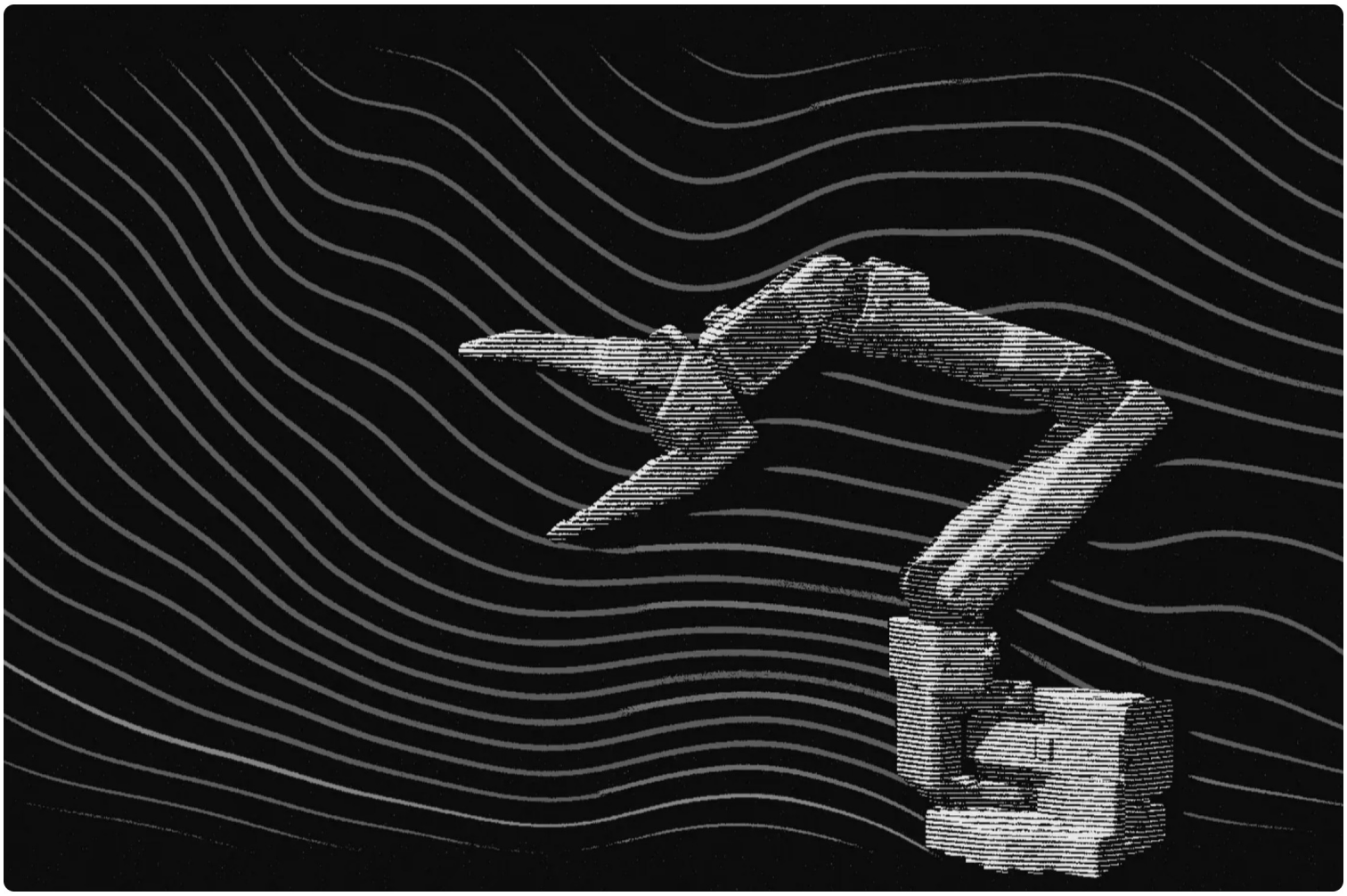




Business & Marketing manages finances, recruitment, and outreach, connecting CUPI with sponsors, new members, and the broader Cornell community.

The Physical Hackathon

CUPI + CEC



The Cornell University Physical Hackathon is for builders who want to make real world impact. Co-hosted by Cornell Physical Intelligence and the Cornell Entrepreneurship Club, it is a regional makeathon where Northeast engineering and entrepreneurship talent builds robots for food, in the field or in the kitchen. Every team gets the same SO-101 robot arm and a software SDK, so the competition comes down to ideas and execution.

DATE
Sat, Nov 14, 2026

LOCATION
Duffield Hall, Cornell

ROBOT
SO-101 arm + SDK

Design week

Teams get seven days to pick a challenge and build their stack against the SDK, testing in simulation before anything touches hardware.

Build day

One day on site at Duffield Hall with 3D printers, extra hardware, and unlimited cloud and AI compute to integrate and test on real robots.

Compete

The robots run live in front of the judges. Simulation-tested stacks meet the real world, and teams compete on the day.

Supported by

Google ANTHROP\C FOUNDATION PICOGRID Kite

For more info, visit hack.cornellphysicalintelligence.com

Why Fund Us?

We give employers access to a very strong and technically able team.

What we're doing

AI is moving off the screen and into the physical world, and most of the hard problems are still unsolved: robots that walk on rough ground, arms that follow plain-language instructions, drones that race through gates with only a camera to go on. This year our members are training a walking policy for an 18-joint metal hexapod, fine-tuning a vision-language-action model for manipulation, and flying an autonomous racing drone in Anduril's AI Grand Prix.

Why it's valuable

Few students get to take a robot from a blank CAD file to a learned policy running on real hardware. Our members do every step themselves: machining and printing parts, designing custom PCBs, writing embedded firmware, and training and deploying models on live sensor data. They develop the full-stack, hands-on experience that robotics, autonomy, and defense companies are looking for.

Who you'll meet

Sponsoring CUPI gives you direct access to some of the most talented and technically able students at Cornell. Our members are self-starters who apply their skills to face unsolved problems.

Our members have worked at

TESLA PICOGRID SPACEX Bloomberg EVERCORE

Where your support goes: Hexapod MKII



Actuators
18× quasi-direct-drive robotics actuators

Structure
CNC plasma cut and bent aluminum

Electronics
Custom board for power distribution and data collection; FPGA for data pre-processing

Compute
NVIDIA DGX Spark for training; Jetson Orin Nano for inference

Sensors
LiDAR, cameras, and 6-axis force sensors in each leg

Software
Isaac Sim and ROS 2

Donor Levels

Every level goes directly toward hardware for Hexapod MKII. Higher levels add more ways to get in front of our members and to work with the team.

Benefits	Gold $5,000	Silver $2,000	Bronze $500
Name and logo on our website	●	●	●
Social media spotlight	●	●	●
Monthly newsletter on our progress	●	●	—
Résumé book access	●	●	—
Logo on posters and merchandise	●	●	—
Logo on the hexapod	●	—	—
Speak at a general body meeting	●	—	—
Recruiting info session for Cornell students	●	—	—
Project collaboration with our team	●	—	—

Current sponsors

CU GEODATA · PICOGRID · onshape · Tantalus · ups

Choose a level

Pick Gold, Silver, or Bronze, or ask us about a custom package.

Email us

Reach out at ab3233@cornell.edu and we'll get everything set up.

Give

Through [DONATION LINK], or in kind: hardware, licenses, or compute at market value.

Cornell Physical Intelligence

Thank you
for your consideration.

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